

- 1) In eukaryotes, general transcription factors usually bind to a promoter sequence called _____
- 2) A small protein tag that marks other proteins for destruction by proteasome is _____
- 3) Enhancers are often located:
- A. directly adjacent to promoter sequence
 - B. only within the coding sequence of a gene
 - C. thousands of base pairs away from the promoter
 - D. on the ribosome
- 4) What is the role of CAP-cAMP complex in the lac operon? This means:
- A. it binds to the operator to block transcription
 - B. it acts as a co-repressor
 - C. it acts as an activator to increase RNA polymerase affinity
 - D. it degrades the repressor protein
- 5) Name the enzyme that adds a methyl group to DNA _____
- 6) Which kind of modification generally leads to loosening of chromatic structure?
- A. DNA Methylation
 - B. Histone Deacetylation
 - C. Histone Acetylation
 - D. RNA Interference
- 7) Alternative RNA splicing allows for: This means:
- A. Faster transcription
 - B. Multiple different proteins to be produced from single gene
 - C. Removal of exons
 - D. Protection of 3' end of the mRNA
- 8) Which of the following stage/process/cycle does the virus genes incorporate into host chromosome?
- A. Reproductive cycle
 - B. Lytic cycle
 - C. Lysogenic cycle
 - D. Prophage cycle
- 9) What is the primary function of siRNA in the cell?
- A. To act as template for protein synthesis
 - B. To facilitate transport of amino acids
 - C. To regulate gene expression by degrading target mRNA
 - D. To serve as primer for DNA replication
- 10) Which of the following is true for locations of the functionally related genes in eukaryotes:
- A. often close to one another with same promoters
 - B. often close to one another with separate promoters
 - C. often distant to one another with same promoters
 - D. often distant to one another with separate promoters